



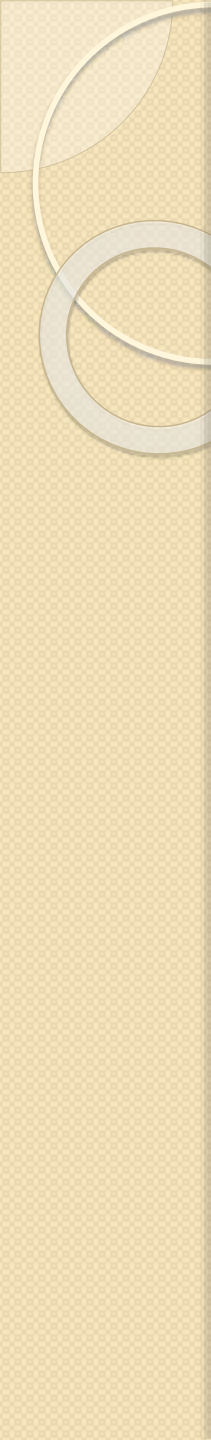
Tech-Edge Lab

Topic: -

**Generative AI & Intelligent Application
Development Bootcamp**

Sub-topic:-

**Industry-Oriented Training Program on
LLMs, AI Tools, AI Agents & Vibe Coding**



Module 1:- AI industry and Careers

Lecture 2:- AI Ecosystem: Understanding AI, ML, Deep Learning, NLP, Computer Vision, Robotics & Generative AI

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1.	AI Ecosystem & Diagram
2.	AI
3.	Types of AI
4.	Machine Learning(ML Workflows) & Types
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6.	Artificial Neural Network (ANN or NN)
7.	ML Vs DL
8.	Natural Language learning(NLP)
9.	NLP task
10.	Computer vision(CV) & its task
11.	Robotics
12.	Gen AI and its working, tradition AI Vs Gen AI
13.	AI Ecosystem Comparison, Real World Example
14.	Future of AI and takeaways

Recap: Introduction to AI World

- **Yesterday We Learned:**
 - Why AI is important
 - AI career opportunities
 - AI vs Traditional Programming
 - Role of AI in software development
- **Today:**
 - How different AI technologies work together.

Activity

Name 3 AI applications you used today.

Name	AI Feature used in that

What is an Ecosystem?

- **Definition**

An ecosystem is a group of technologies that work together to create intelligent systems.

- **Human Intelligence Example:**

- Human uses:
- Eyes → Seeing
- Ears → Listening
- Brain → Thinking
- Memory → Learning
- Decision Making → Action

- AI systems also combine multiple technologies.

Artificial Intelligence

```
graph TD; AI[Artificial Intelligence] --> Dashed1[ ]; Dashed1 --> ML[Machine Learning]; Dashed1 --> RBAI[Rule Based AI]; ML --> DL[Deep Learning]; DL --> Dashed2[ ]; Dashed2 --> NLP[NLP]; Dashed2 --> CV[Computer Vision]; NLP --> GenAI[Generative AI]; GenAI --> AIA[AI Applications]; AIA --> Robotics[Robotics];
```

Machine Learning

Rule Based AI

Deep Learning

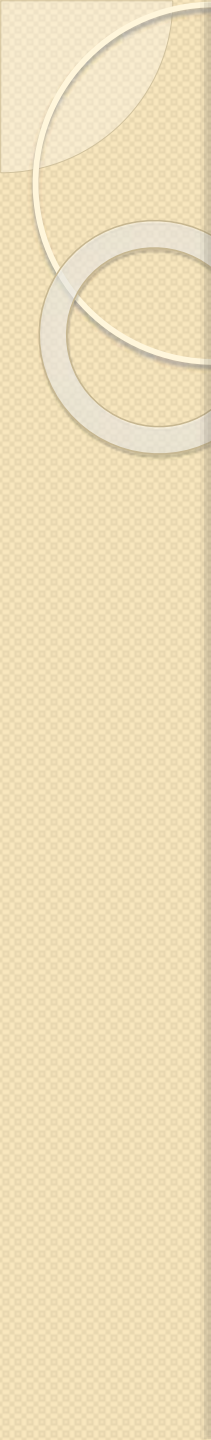
NLP

Computer Vision

Generative AI

AI Applications

Robotics



Important Points

AI is the biggest field.

ML is a technique to achieve AI.

Deep Learning is a powerful ML method.

NLP helps machines understand language.

Computer Vision helps machines see.

Generative AI creates new content.

Robotics combines AI with machines.

- Human:
 - **Parts** → **Field of AI**
 - Eyes → ?
 - Brain → ?
 - Memory → ?
 - Learning → ?
 - Language → ?
 - Decision → ?

Answers

- Eyes → Computer Vision
- Brain → AI
- Memory → Data
- Learning → Machine Learning
- Language → NLP
- Decision → AI Algorithms

Activity 1

- **Identify the Technology**
students:

1) Face Unlock

- Answer:
?

2) Google Translate

- Answer:
?

3) ChatGPT

- Answer:
?

4) Self-driving Car

- Answer:
?

Answers

- **Face Unlock**

Answer:

Computer Vision + Deep Learning

- **Google Translate**

Answer:

NLP

- **ChatGPT**

Answer:

NLP + Deep Learning + Generative AI

- **Self-driving Car**

Answer:

Computer Vision + ML + Robotics

Artificial Intelligence (AI)

- **Definition**

Artificial Intelligence enables machines to perform tasks that normally require human intelligence.

- **AI Capabilities**

- Learning
- Reasoning
- Planning
- Decision Making
- Problem Solving

- **Examples**

- Google Maps
- Siri
- Alexa
- ChatGPT
- Recommendation Systems

- **Activity 2**

- Is it AI or Not?**

- Example AI? (Yes/No)

- Calculator

- Google Maps

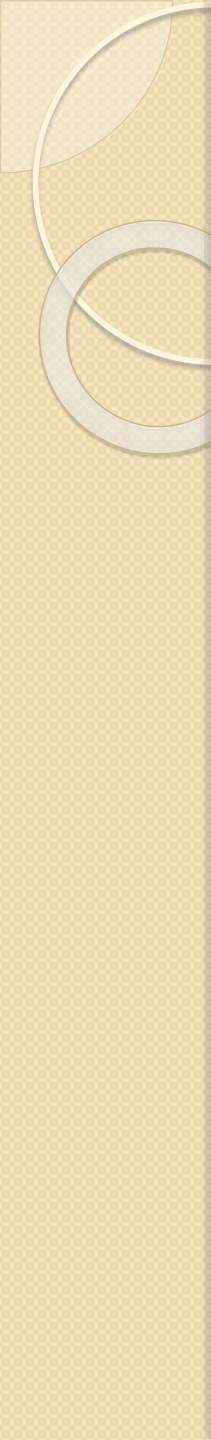
- YouTube Recommendation

- Simple ATM Machine

- ChatGPT

- Snapchat Filters

- TV Remote control



Example	Is It AI ?
Calculator	No
Google Aps	Yes
You tube recommendation	Yes
Simple ATM machine	No
Chat Gpt	Yes
Snapchat Filters	Yes
TV Remote Control	No

Types of AI

1. Artificial Narrow Intelligence (ANI):- Artificial Narrow Intelligence (ANI), or weak AI, refers to AI systems designed to excel at a single, highly specific task.

Examples:

Most AI today.

ChatGPT

Google Translate

Alexa

2. Artificial General Intelligence (AGI)

AI system which develop Human-level intelligence.

Not achieved yet.

3. Artificial Super Intelligence (ASI)

AI beyond human intelligence.

Future concept.

Machine Learning

Definition

Machine Learning allows computers to learn from data without being explicitly programmed.

Traditional Programming

Rules + Data (C tokens, function, keywords)

Program

Output (sum.c , cal.c)

- **Machine Learning**

Data + Output Examples (C tokens, function, keywords + examples)

ML Model (supervised, unsupervised, reinforcement)

Learn Rules

Example: Spam Detection

Traditional Approach:

Create rules:

- Contains "free money"
- Unknown sender
- Suspicious links

Problem:

* Too many rules.

Machine Learning:-

- Give thousands of emails.
- AI learns patterns.

Types of Machine Learning

1. Supervised learning
2. Unsupervised learning
3. Reinforcement learning'

Supervised Learning

Learning with labelled data.

Examples:

- Spam Detection
- Disease Prediction
- Student Result Prediction

Supervised learning is exactly like studying for an exam using a practice quiz that includes the answer key.

Unsupervised Learning

Finding hidden patterns.

Examples:

Customer Segmentation

Shopping Behavior

- Unsupervised learning is like walking into a massive library where none of the books have covers, titles, or genres written on them—and there is no librarian or answer key to help you.
- Instead of checking an answer key, the AI must explore the books on its own, find similarities in how they are written, and automatically group them together (e.g., grouping books with magic spells into one pile, and books with spaceships into another).

Reinforcement Learning

Learning through rewards.

Examples:

Game AI

Robots

Self-driving cars

- Reinforcement learning is exactly like training a puppy to do a trick using treats and time-outs.
- The AI does not have an answer key, and it does not have a teacher telling it what to do. Instead, it is dropped into an environment and forced to learn through trial and error. When it does something right, it gets a "reward" (a digital treat). When it crashes or fails, it gets a "punishment" (a penalty score).

Activity 3

Q- Which ML type is used?

1) Predict house price

Answer:

?

2) Group customers based on shopping

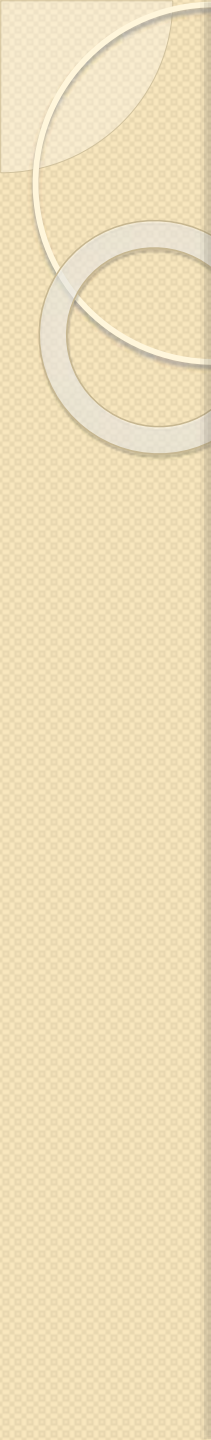
Answer:

?

3) Robot learning walking

Answer:

?



Answers:-

Predict house price

Answer:

Supervised

Group customers based on shopping

Answer:

Unsupervised

Robot learning walking

Answer:

Reinforcement

Machine Learning Workflow

Collect Data



Clean Data



Train Model



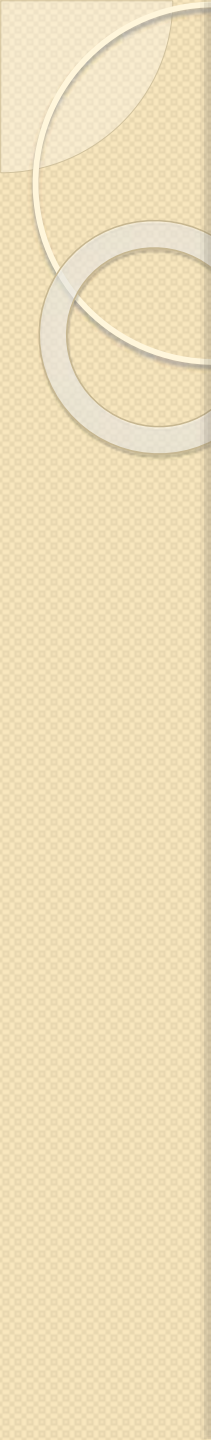
Test Model



Deploy



Prediction



Example:

Student Placement Prediction

Input:

CGPA

Skills

Attendance

Projects

Output:

Placed / Not Placed

Deep Learning

Definition

Deep Learning is a subset of Machine Learning that uses artificial neural networks.

Inspired by human brain.

Used for:

- Images
- Voice
- Video
- Language

Examples:

- Face Recognition
- ChatGPT
- Image Generation

Artificial Neural Network

Human Brain:

Neuron → Connection → Learning

AI:

Input Layer



Hidden Layers



Output Layer

More layers = Deep Learning

Human Brain

VS.

Artificial Neural Network

Different in form, Similar in Function

Neurons

Basic processing units of the brain

Synapses

Connections between neurons to transmit signals

Learning

From experience and feedback

Memory

Stores knowledge and experiences

Adaptation

Adjusts and evolves over time



SIMILARITIES



Information Processing

Both process information



Learning

Both learn from data/experience



Memory

Both store knowledge



Adaptation

Both improve over time

Nodes

Basic processing units (like neurons)

Weights

Connections between nodes (like synapses)

Learning

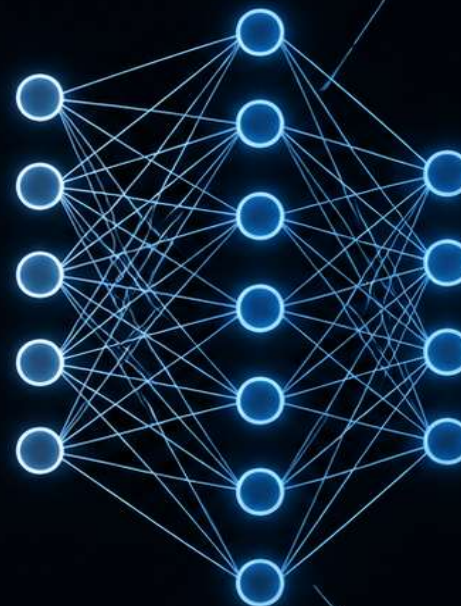
From training data

Memory

Stored in weights and biases

Adaptation

Improves with more data



HUMAN BRAIN

~86 billion neurons

Synapses are chemical

Highly parallel

Works with very low power (~20W)

Learns from limited data and generalizes well

Emotions, consciousness, creativity



ARTIFICIAL NEURAL NETWORK

Millions to billions of artificial neurons

Connections are numerical (weights)

Parallel but limited by hardware

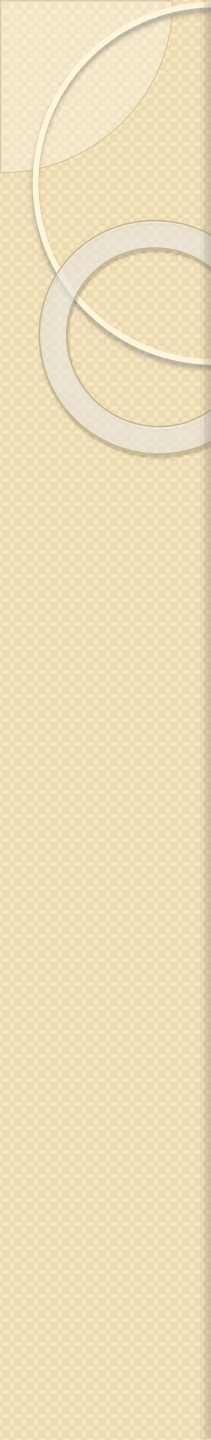
Requires high power (GPUs/TPUs)

Needs large data, less generalization

No emotions or consciousness (currently)



Inspiration Biologically Inspired AI



Activity 4

How does Face Recognition work?